

Cloud Architecture of AeroLink Airlines

(Case Study)

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Abstract: This case study presents a comprehensive cloud architecture designed for AeroLink Airways, a global airline focused on innovation, customer experience, and operational excellence. The architecture leverages Microsoft Azure services to address the challenges of data silos, real-time analytics, and machine learning capabilities. It follows a structured approach starting from the discovery phase to implementation, with a strong emphasis on scalability, efficiency, and compliance.

The architecture employs a multi-layered data processing approach—**Bronze (Raw Data)**, **Silver (Curated Data)**, and **Gold (Aggregated Business Data)**—supported by services such as **Azure Data Lake**, **Synapse Analytics**, **Stream Analytics**, **Delta Lake**, and **Azure Data Factory**. It enables the ingestion, transformation, and aggregation of data from IoT sensors, booking systems, APIs, and vendor platforms. Further, the architecture supports real-time analytics, AI/ML model integration, and business intelligence reporting through dashboards and geospatial insights.

This article explores the design, data flow, key components, and real-world applications of the architecture, offering a scalable, flexible, and intelligent framework for modern enterprise data solutions on Azure.

Introduction

In today's data-driven world, the aviation industry relies heavily on real-time insights and integrated systems to ensure efficient operations and an exceptional customer experience. This case study focuses on AeroLink Airways, a representative airline that demands a modern, cloud-native solution to manage diverse data sources and support its analytical and operational needs.

As data engineers, our objective was to design a robust and scalable cloud architecture using **Microsoft Azure** that could address AeroLink's challenges—ranging from fragmented data sources and inefficient reporting to the need for real-time insights and cost optimization. The architecture presented here aligns with best practices in cloud design and showcases how cloud services can be orchestrated to power intelligent decision-making in the airline industry.

In this article, we'll walk through a **Lakehouse architecture** built on **Microsoft Azure**, which unifies the power of **data lakes** and **data warehouses**. This architecture uses services like Azure Synapse Analytics, Delta Lake, Azure Data Factory, Stream Analytics, and more to handle **real-time and batch data processing**, **data transformation**, and **AI integration**.

Objectives

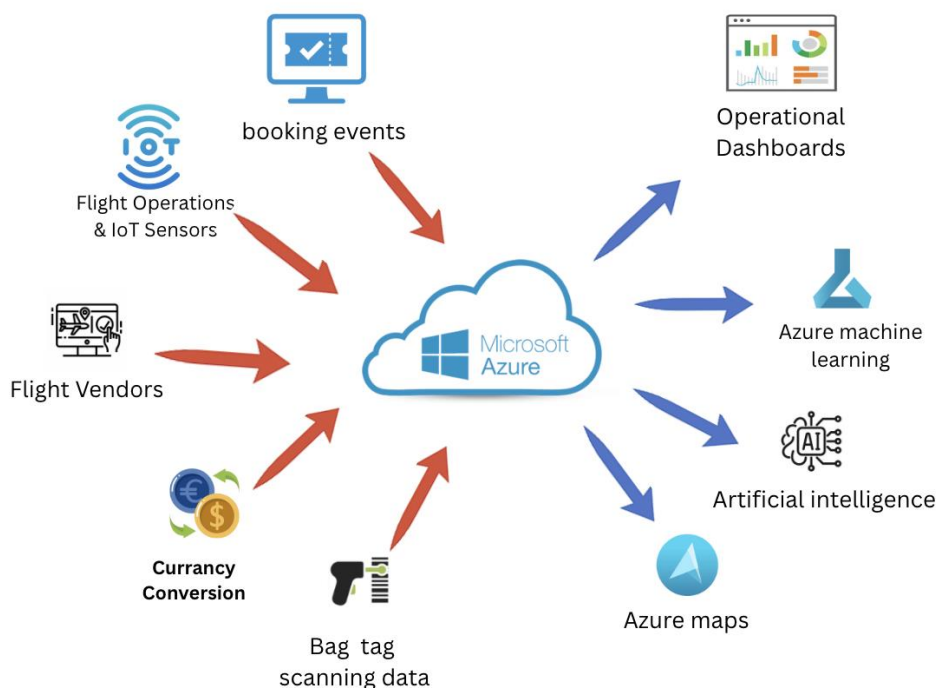
- **Cost Optimization & Performance Scalability:** Optimize costs while ensuring high-performance data processing that scales with demand.
- **Real-Time Data Processing for Operational Efficiency:** Enable real-time analytics for flight scheduling, flight updates baggage tracking, and management.
- **AI-Powered Decision Intelligence for Airline Optimization:** to optimize flight scheduling, dynamic

Discovery Phase : Understanding the Problem and Business Requirements

The discovery phase is the foundation of any successful cloud data architecture. Before designing the solution, we identified key business needs and technical requirements through stakeholder meetings, data audits, and workflow analysis. The objectives were:

- Integrate multiple data sources (IoT sensors, APIs, booking systems).
- Handle real-time and batch data streams efficiently.
- Provide a scalable and cost-effective storage solution.
- Enable analytics, dashboards, and machine learning.
- Deliver curated and aggregated data for decision-making.

During this phase, we also engaged with stakeholders to uncover key business challenges, user expectations, and existing data pain points. One of the most important tasks was identifying all **data sources** and defining the **intended destinations** of processed data.



Identified Data Sources:

- **IoT Sensors** – High-velocity telemetry data from aviation devices.
- **Booking Events** – Data from customer booking systems.
- **Flight Vendor Systems** – Third-party APIs for flight schedules and data feeds.
- **Currency Conversion Services** – External APIs providing real-time exchange rates.
- **Bag Tag Scanning Devices and Cargo data**– Edge devices capturing scanned data across airports.

Identified Data Destination:

- **Power BI Dashboards and Reports** – For executive and operational reporting.
- **Azure Machine Learning Services** – For predictive analytics and AI use cases.
- **Azure Maps** – For geospatial visualization.

By identifying **where data comes from** and **where it needs to go**, we laid the groundwork for a data architecture that is efficient, scalable, and aligned with business outcomes.

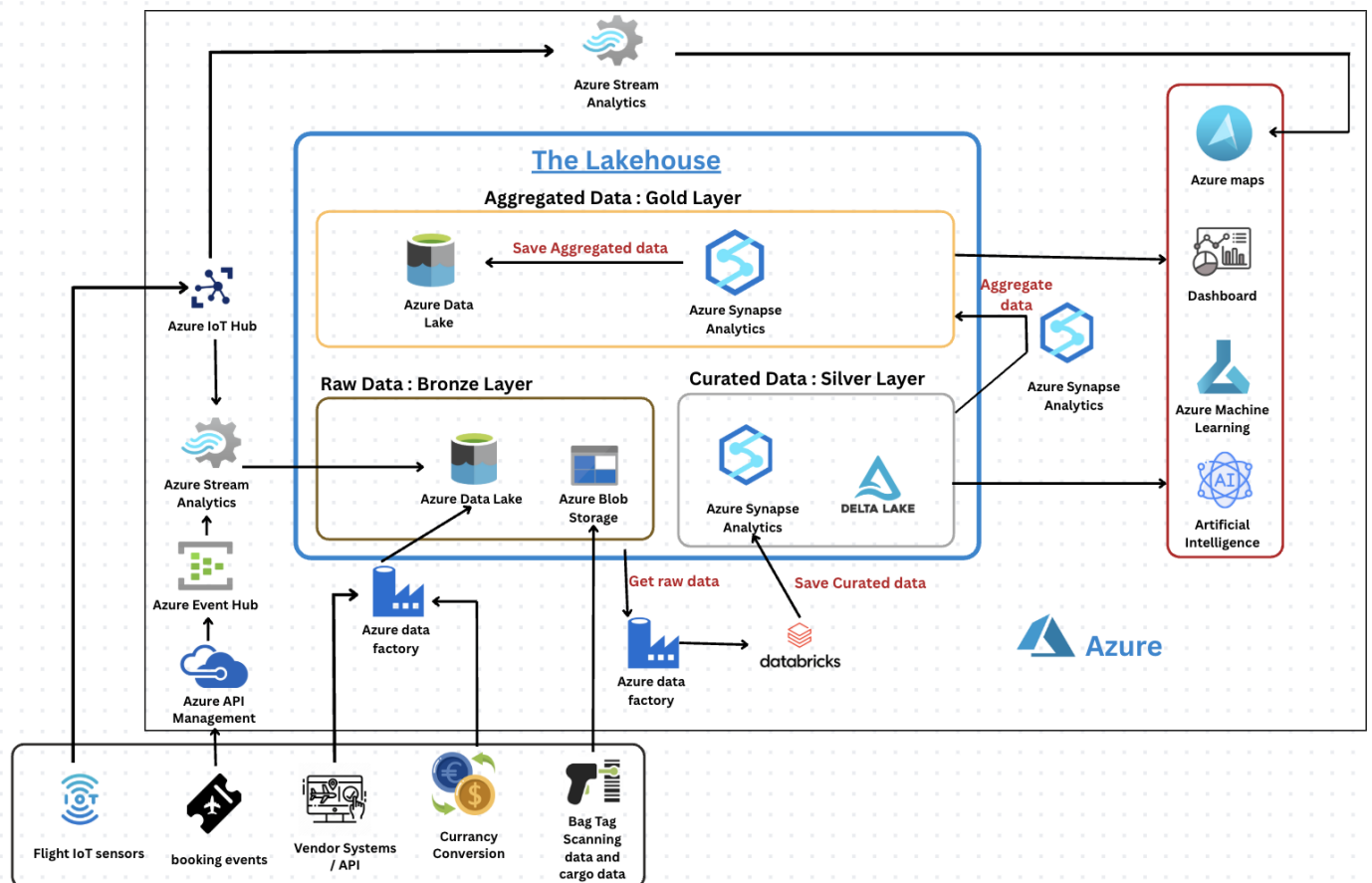
Proposed Cloud Architecture Overview

The AeroLink Airways cloud solution is built on the principles of the **Lakehouse Architecture**, which unifies the strengths of traditional data lakes and data warehouses. This approach enables efficient storage of raw and semi-structured data while supporting advanced analytics and structured queries—all within a single, scalable architecture.

The architecture is organized using the medallion architecture model, comprising three logical layers:

- **Bronze Layer (Raw Data Storage):** Ingests real-time and batch data from multiple sources via Azure Event Hub, Azure IoT Hub, and Azure Data Factory. This layer stores unprocessed data in Azure Data Lake Storage Gen2 and Azure Blob Storage, serving as the single source of truth for raw data.
- **Silver Layer (Cleansed & Enriched Data):** Data engineers use Azure Synapse Analytics to clean, validate, and join datasets from the Bronze layer. Business rules and transformations are applied here to convert raw data into refined, analytics-ready formats.
- **Gold Layer (Aggregated & Curated Data):** This layer houses the aggregated and business-consumable datasets. It supports high-performance analytics, machine learning models, and Power BI dashboards. Azure Synapse SQL pools or dedicated SQL endpoints serve curated data with high reliability.

This Lakehouse-driven architecture provides flexibility for AeroLink to handle diverse data types (structured, semi-structured, unstructured) and enables rapid data exploration, governance, and innovation.



Components Involved

- **Azure Event Hub** – Real-time ingestion from booking systems, IoT sensors, and operations
- **Azure IoT Hub** – Data from aircraft sensors and IoT devices
- **Azure Data Factory** – Orchestrates batch data movements and transformations
- **Azure Data Lake Storage Gen2** – Raw and intermediate storage
- **Azure Blob Storage** – Staging and archiving
- **Azure Synapse Analytics** – For querying, transformation, and analytical workloads
- **Power BI** – For interactive dashboards and reporting
- **Azure API Management** – Manages internal and external APIs securely and allows controlled access to services like currency conversion and vendor integrations.
- **Azure Maps** – Provides geographic intelligence, including mapping of flight routes and geolocation analytics.
- **Azure Stream Analytics** – Processes streaming data in real-time before routing it to the lakehouse.
- **Azure Machine Learning** – For predictive analytics, including flight delay predictions, customer segmentation, etc.

Benefits of the Architecture

- **Unified Storage & Analytics** – Lakehouse design integrates flexibility of lakes with the performance of warehouses
- **Real-time Capabilities** – Near real-time processing for time-sensitive operations
- **Cost-Optimized** – Pay-as-you-go model ensures financial efficiency

Data Flow and Processing Logic:

Ingestion:

Data Types and Ingestion Strategy

In this case study, we deal with **two types of data**:

1. Real-time Data

- **Flight IoT Sensors**: These generate telemetry including aircraft health metrics and live geospatial data (e.g., current flight location).
- **Real-time Booking Events**: Generated from user booking activities.

2. Batch Data

- **Vendor System Data**: Includes third-party partners such as fuel suppliers, maintenance teams, and ticket resellers.
- **Currency Conversion APIs**
- **Bag Tag Scanning and Cargo Data**

Real-time Data Ingestion

Flight IoT Sensor Data

To ingest telemetry from aircraft, we use **Azure IoT Hub** — the primary gateway for real-time IoT data. This data is split into two flows:

- **Geospatial Data:** Sent to **Azure Stream Analytics**, then forwarded to **Azure Maps** to visualize the live location of flights.
- **Telemetry Data:** Streamed through **Stream Analytics** and stored in **Azure Data Lake** for advanced analytics and machine learning, such as **predictive maintenance**.

Alternative Storage Options Considered:

- **Azure Blob Storage:** Less optimized for analytics workloads.
- **SQL Database:** Not suitable for large-scale time-series IoT data.
- **Cosmos DB:** High cost at scale.
- **Time Series Insights:** Not ideal for long-term storage needs.

Given the need for analytics readiness and scalability, **Azure Data Lake** is the most suitable option.

Booking Event Data

Booking events are real-time and need to be handled with scalability and security:

1. Data is first received via **Azure API Management** to ensure secure and scalable API access.
 - *Note:* Alternatives like **Logic Apps** were considered, but they are not ideal for high-throughput, real-time APIs.
2. **API Management** forwards the data to **Azure Event Hub**, which offers low-latency, high-throughput ingestion with durability.
 - *Alternative:* **Azure Service Bus** was not selected, as it is more suited for messaging queues rather than real-time event streaming.
3. **Stream Analytics** then processes this booking data and stores it in **Azure Data Lake** for further processing.

Batch Data Ingestion

Vendor Systems & Currency Conversion APIs

The airline collaborates with multiple third-party vendors. Data from these systems and APIs is extracted using **Azure Data Factory** and stored in **Azure Data Lake** — the centralized raw zone for batch data.

Bag Tag Scanning and Cargo Data

Bag scanning is performed during check-in and loading processes. Since there's **no immediate need** for real-time updates, this data is processed in **batches**.

- The scanned tag data is collected via HTTP and directly stored in **Azure Blob Storage**.

- **Why Blob Storage?**

It is cost-effective and sufficient here since the data:

- Doesn't require hierarchical structure.
- Doesn't need immediate transformation.
- Doesn't demand deep analytical processing on ingestion.

This batch approach also improves **data accuracy**, avoiding the noise and inconsistencies that can come with real-time data.

Curated Layer (Silver Layer)

After data is ingested and stored in the **Bronze Layer** as raw data, it moves into the **Curated Layer**, also known as the **Silver Layer**. This layer focuses on **data transformation, enrichment, and integration** — preparing it for analytical use and machine learning applications.

Purpose:

- Clean, transform, and integrate raw data from multiple sources
- Prepare high-quality datasets for AI, ML, and advanced analytics
- Ensure consistency and usability of data for downstream systems

Process and Tools:

- **Azure Data Factory:** Orchestrates and automates the data movement from the raw zone to transformation layers.
- **Azure Databricks:** Performs complex transformation logic using Apache Spark. This includes:
 - Removing duplicates and nulls
 - Converting formats (e.g., JSON to Parquet)
 - Joining datasets (e.g., IoT data with booking info)
 - Feature extraction for machine learning

Storage:

- **Delta Lake on Azure Data Lake** is used for storing curated data:
 - Offers ACID compliance and time travel
 - Supports schema evolution and is optimized for ML pipelines

Aggregated Layer (Gold Layer)

Once data in the Silver Layer is refined and curated, it is further **aggregated and modeled** to support high-performance reporting, KPI dashboards, and business intelligence insights — this stage is known as the **Gold Layer**.

Purpose:

- Aggregate and summarize curated data
- Build data models and pre-calculated metrics
- Support business reporting tools such as Power BI

Process and Tools:

- **Azure Synapse Analytics (SQL Pools):**
 - Aggregates curated datasets using SQL queries
 - Performs data modeling (fact/dimension tables, star/snowflake schema)
 - Optimized for fast querying and dashboard integration
- **Azure Data Lake** is also used in parallel to store aggregated datasets in Delta format for auditability, ML use cases, and external reporting needs.

Consumption:

- **Power BI** directly connects to **Azure Synapse SQL Pools** and/or **Delta Lake** to provide real-time dashboards, trend analysis, and interactive visualizations to business users.

Why This Layered Approach?

- **Silver Layer (Curated Layer):**
 - **Description:** Contains cleaned, transformed, and integrated data from the raw (bronze) layer.
 - **Storage:** Stored in Delta Lake on top of Azure Data Lake.
 - **Key Benefits:** Ensures consistent, high-quality, and structured data that's ready for analytics and machine learning workloads.
- **Gold Layer (Aggregated Layer):**
 - **Description:** Includes aggregated, summarized, and business-ready data models.
 - **Storage:** Stored in Azure Synapse SQL Pools and optionally in Delta Lake for additional use cases.
 - **Key Benefits:** Optimized for reporting, executive dashboards, and business intelligence tools like Power BI.

Pipeline Failure Handling Strategy

1. Azure IoT Hub & Azure Stream Analytics (Real-time Data Ingestion)

- **Failure Scenario:** Data not flowing from sensors or events.
- **Strategy: Retry with Timeout**
 - Use **Stream Analytics retry policies** to reconnect to input sources.
 - Set **maximum retries** and use **DLQ (Dead Letter Queue)** to catch unprocessed events.
 - Enable **checkpoints** so processing can resume from the last known good state.
 - Alert team if ingestion is down for a set time window (timeout).

2. Azure Data Factory (Data Movement & ETL)

- **Failure Scenario:** ADF pipeline fails due to data format issues, missing files, or service unavailability.
- **Strategy: Retry with Timeout**
 - Configure **retry policies** in each pipeline activity (max retry count and interval).
 - Use **timeout settings** to avoid long-running, hanging jobs.
 - Enable **error output** in copy activity to redirect failed records to a separate location for later inspection.

3. Azure Databricks (Data Processing & Curating)

- **Failure Scenario:** Databricks job fails due to bad data, script errors, or compute issues.
- **Strategy: Timeout + Alert**
 - Use **job timeout settings** in Databricks.
 - Integrate **Azure Logic Apps or Azure Monitor Alerts** to notify devs on failure.
 - Optionally, add logic to **resubmit the job from where it failed** using checkpoints or job run IDs.

4. Azure Synapse Analytics (SQL Pool & Aggregations)

- **Failure Scenario:** Query timeout, resource contention, or bad data joins.
- **Strategy: Retry (for transient issues) + Timeout**
 - Use **stored procedure-based retry logic** for transient errors.
 - Use **timeouts for long-running queries** to prevent system lock.
 - Log failures to Azure Monitor and retry intelligently depending on error code.

5. Delta Lake (Curated & Gold Layer Storage)

- **Failure Scenario:** ACID transaction fails or data corruption during write.
- **Strategy: Retry + Use Delta Transaction Logs**
 - Retry failed writes using **Databricks job reruns**.
 - Delta Lake has **built-in transaction logs** that help **restore or rerun** the last successful version.
 - Validate data post-write using **record counts/checksum**.

6. Power BI Dashboard / AI & ML Use Cases

- **Failure Scenario:** Data not available or outdated.
- **Strategy: Timeout + Alert**
 - Alert users if data hasn't refreshed in the expected interval.
 - Use **dataflow or dataset refresh settings with retries**.

Monitor data freshness using **Azure Synapse and Data Factory triggers**.

Conclusion

Designing a modern cloud architecture for AeroLink Airlines using Microsoft Azure demonstrates how cloud-native solutions can revolutionize data management in the aviation industry. By implementing a Lakehouse approach, we successfully unified real-time and batch data processing, ensured scalable storage, and enabled advanced analytics across the organization.

From ingesting high-velocity IoT sensor data to transforming it into meaningful insights through curated and aggregated layers, this architecture supports not only operational efficiency but also data-driven decision-making. Azure services like Data Factory, Synapse Analytics, Delta Lake, Stream Analytics, and Power BI work seamlessly to deliver a resilient and intelligent data ecosystem.

This architecture sets the foundation for future innovations in AI, predictive maintenance, and personalized customer experiences—making AeroLink not just an airline, but a data-enabled enterprise. As aviation continues to evolve, cloud platforms like Azure will remain pivotal in shaping the industry's digital transformation.